KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering | SAP BTP Developer Project

Hire-to-Retire (H2R)

Employee Lifecycle Management in SAP HR

Attribute	Details
Process Name	Hire-to-Retire (H2R) — Employee Lifecycle Management in SAP HR
SAP System	SAP S/4HANA 2023 (On-Premise) + SAP SuccessFactors (Cloud Hybrid)
Primary Modules	HCM: PA, OM, TM, PY-IN, ESS/MSS | SF: EC, Recruit, Onboard, LMS, PMG
BTP Services	CAP (Node.js), Integration Suite (CPI), HANA Cloud, Build Apps, Workflow Mgmt., Alert Notification, XSUAA / SAP IAS
Compliance	EPF Act, ESI Act, Shops & Establishments Act, Income Tax Act (TDS Sec. 192), Payment of Gratuity Act 1972, Maternity Benefit Act 2017
Scope	SAP HCM: PA · OM · TM · PY-IN · ESS/MSS | SuccessFactors: EC · Recruit · Onboard · LMS · PMG | BTP: CAP · Integration Suite · HANA Cloud · Build Apps · Workflow

TOPIC:- Hire-to-Retire (H2R) — Employee Lifecycle in SAP HR

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"From the first offer letter to the last exit interview — SAP HR connects every touchpoint of the employee journey."

1. Problem Statement

Organizations managing large workforces often struggle with fragmented, manual processes spanning recruitment, onboarding, payroll, performance management, and exit. These disconnected silos create data redundancy, payroll inaccuracies, statutory non-compliance, and a poor employee experience. Without a unified system, HR teams spend most of their time on administrative burdens rather than strategic talent initiatives.

Core Challenge: How can an organization implement a seamless, integrated, and automated employee lifecycle system — from the moment a candidate is hired until they retire or exit — using SAP HR (HCM) as the backbone?

Key Pain Points Addressed:

- Lack of real-time visibility into headcount, attrition, and workforce analytics
- Delayed or error-prone payroll due to manual data entry across disconnected systems
- Non-compliance with statutory requirements (PF, ESI, TDS, Gratuity) due to siloed data
- Inefficient onboarding causing high early-exit rates among new joiners
- Absence of structured career and succession planning tools for long-term retention
- Cumbersome and delayed separation processes causing poor employee exit experience

2. SAP HR & H2R Framework Overview

SAP Human Capital Management (HCM) is the enterprise module covering the full spectrum of HR functions. The Hire-to-Retire (H2R) process maps every employee event — from hiring to retirement — into integrated, automated transactions within SAP.

Core Module	SAP HCM (Human Capital Management)
Sub-Modules	PA · OM · Payroll · Time Mgmt · ESS/MSS · LSO
Integration	SAP FI, SAP MM, SAP SuccessFactors
Primary Infotypes	IT0000 – IT9999 (Personnel Actions, Pay, Time, Benefits)
Key T-Codes	PA40, PA30, PA20, PO10, PO13, PC00_M40_CALC, PT60
Reporting Tools	SAP BI/BW, Ad-hoc Query, HR Analytics Cloud

3. Solution — Step-by-Step H2R Process

The H2R lifecycle in SAP HR is structured into seven sequential phases. Each phase is driven by specific transactions, infotypes, and IMG configuration steps:

Phase	T-Code	Key Activity	Output
Phase 1	**PO10 / PPME**	Enterprise Structure: Define Company Code, Pers. Area, Pers. Sub-Area, Emp. Group/Sub-Group, Cost Centre	Org Chart, Cost Centre
Phase 2	**PA40 / PA30**	Hiring Action: Maintain IT0000 (Actions), IT0001 (Org Assignment), IT0002 (Personal Data), IT0006 (Address), IT0007 (Work Schedule)	Employee Master Record
Phase 3	**PA30 / IT0008-9**	Onboarding: Assign Job, Position, Pay Scale; configure Basic Pay (IT0008), Bank Details (IT0009), Emergency Contact (IT0021)	Payroll Enrollment, Offer Letter
Phase 4	**PT60 / CATS**	Time Management: Time Evaluation, Attendance/Absence Types, Work Schedules, Leave Quotas (IT2001, IT2002, IT2006)	Attendance Records, Leave Balances
Phase 5	**PC00_M40_CALC**	Payroll Run: India Payroll driver; PF/ESI/TDS statutory deductions; payslip generation; bank transfer advice	Payslip, Bank Transfer
Phase 6	**LSO / PHAP_SEARCH**	Performance & Training: Appraisals, KRA tracking, Learning Solution, Succession Planning, Career Dev. Plans	KRA Reports, Certificates
Phase 7	**PA40 (Termination)**	Separation: Initiate Termination Action; Final Settlement (IT0014); generate Experience Letter & F&F; Statement	F&F; Settlement, Exit Cert.

3.1 Enterprise Structure & Personnel Administration

Before initiating any H2R transaction, the enterprise structure is configured in SAP SPRO (IMG). The hierarchy flows from Client → Company Code → Controlling Area → Personnel Area → Personnel Sub-Area → Employee Group → Employee Sub-Group. Organizational Units are created via PO10, Positions via PO13, and Jobs via PO03. Cost Centres (KS01) link Org Units to Finance for budget control.

3.2 Payroll Configuration — India Specific

- Pay Scale Groups & Levels configured via T-Codes OH11 / OH12; Wage Types assigned per Employee Sub-Group
- Statutory Deductions: Provident Fund (PF), Employees' State Insurance (ESI), Professional Tax (PT), TDS via Feature TARIF
- Payroll Driver: PC00_M40_CALC | Log Review: PC00_M40_CEDT | FI Posting: PC00_M40_CIPE
- Payslip via HRFORMS / Adobe Forms; Bank Transfer Advice via T-Code PREC; Off-cycle for bonus/arrears

4. Technology Stack & Integration Architecture

Layer	Component	Purpose
Core ERP	**SAP ECC 6.0 / S/4HANA**	Master H2R platform
HR Module	**SAP HCM (PA,OM,PY,TM)**	All HR sub-modules
Database	**SAP HANA / MaxDB**	In-memory processing
Frontend	**SAP GUI / Fiori**	Desktop & mobile UI

Layer	Component	Purpose
Integration	**BAPIs / IDocs / RFC**	Cross-module data flow
Reporting	**SAP BI/BW, Ad-hoc**	HR KPIs & analytics
Self-Servic e	**ESS / MSS Portal**	Leave, claims, updates
Compliance	**India Payroll MX0**	PF/ESI/TDS/Gratuity

4a. SAP H2R Cross-Module Integration

SAP HR integrates with multiple enterprise modules and external systems for seamless data flow:

Integration	Module	Data Exchange	Method
HR → Finance	**SAP FI/CO**	Payroll G/L postings; cost centre debit	RPCIPE00/IDoc
HR → Materials	**SAP MM**	Employee asset & uniform procurement	BAPI / RFC
HR → Cloud	**SuccessFactors**	Talent, recruiting & learning data sync	SAP CPI/MID
HR → IT Security	**LDAP / AD**	Auto user-ID creation on hire; disable on exit	IDM / BAPI
HR → Bank	**External Bank**	Salary disbursement via NEFT / ECS	DME / PREC

4b. Business Impact & KPIs

40%
Fewer payroll errors post-automation

60%
Faster onboarding via infotype workflow

100%
Statutory compliance — PF/ESI/TDS

30%
HR admin time saved via ESS/MSS

Business Area	Before SAP H2R	After SAP H2R	Impact
Payroll Processing	5–7 days, manual	**Same-day automated run**	**~80% faster**
Employee Onboarding	2–3 weeks paperwork	**3–5 days digital workflow**	**~65% faster**
Leave Management	Manual registers/email	**Real-time ESS portal**	**Zero manual**
Statutory Filing	Error-prone, penalties	**Auto-deduction & filing**	**100% compliant**
HR Reporting	Weekly manual MIS	**Real-time BI dashboards**	**Instant insights**

5. Unique Points & Key Differentiators

What Makes SAP H2R Implementation Distinctive
■ Single Source of Truth: All employee data from hire to retire centrally maintained — eliminates reconciliation across spreadsheets and siloed HR systems.
■ Infotype Architecture: SAP's time-bound Infotype design (IT0000–IT9999) ensures full audit trails with validity start/end dates for every data record.
■ Real-Time Payroll Integration: Time Management data flows directly into Payroll without manual intervention — reducing errors and ensuring accurate, on-time payslips.
■ Statutory Compliance by Design: India Payroll schema PC00_M40 has built-in logic for PF, ESI, PT, TDS, and Gratuity — auto-calculating per current regulatory thresholds.
■ Configurable Workflow Engine: PA40 Action Menus are fully configurable — custom hiring, transfer, promotion, and termination workflows without any code changes.
■ Role-Based Security: SAP authorization objects enforce HR data privacy — payroll data restricted to payroll admins; managers see only their team via MSS.

6. Future Improvements & Roadmap

While the current H2R blueprint delivers comprehensive lifecycle management, the following enhancements are planned for future phases:

- **SAP S/4HANA Migration:** Upgrade from ECC 6.0 to S/4HANA for real-time HR analytics, simplified data model, and Fiori-native UX — eliminating legacy GUI transactions.
- **SAP SuccessFactors Integration:** Connect on-premise HCM with SuccessFactors cloud modules (Recruiting, LMS, Performance & Goals) for a full cloud-hybrid H2R experience.
- **AI-Powered Talent Analytics:** Integrate SAP Analytics Cloud (SAC) with machine learning to predict attrition, identify skill gaps, and proactively recommend training programs.
- **Conversational ESS (Chatbot):** Deploy SAP Conversational AI for leave applications, payslip queries, and HR requests via WhatsApp or Microsoft Teams.
- **Blockchain Credential Verification:** Use blockchain for tamper-proof offer letters, experience certificates, and educational credential verification to eliminate fraud.

Conclusion

This project provided a comprehensive overview of the Hire-to-Retire (H2R) lifecycle in SAP HR, demonstrating how integrated modules streamline key HR functions such as recruitment, onboarding, payroll, performance management, and employee separation. By utilizing SAP HCM's infotype-based structure and automated workflows, organizations can minimize manual errors, ensure statutory compliance, and maintain accurate, real-time employee data.

Overall, the H2R framework transforms traditional HR operations into a more efficient and strategic function. It enhances decision-making through better visibility and analytics while improving employee experience across the lifecycle. Thus, SAP HR serves as a powerful tool for organizations aiming for operational excellence and long-term workforce management.

The Hire-to-Retire lifecycle in SAP HR transforms HR from an administrative function into a strategic enabler — connecting every touchpoint of the employee journey within a single, integrated enterprise platform.